IOE LAB PROJECT REPORT

ON

FIRE ALARM SYSTEM WITH SIREN AND WATER SPRINKLER

Submitted in fulfillment of the requirement of University of Mumbai

For the degree of

Bachelor of Engineering
(Information Technology)

By

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Under the Guidance of

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Knowledge to Empower

DEPARTMENT OF INFORMATION TECHNOLOGY ENGINEERING

TERNA ENGINEERING

COLLEGEUNIVERSITYOF MUMBAI

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PROJECT REPORT APPROVAL FOR B. E.

The Project Report entitled **"Fire Alarm System with Siren and Water Sprinkler ",** Submitted by **"Mohd Salman" (TU4S2021018), "Umaid Shaikh" (TU4S2021005), "Hrushikesh Dunde" (TU4S2021008) and "Vivek Shirsat" (TU4F1819070)"** is approved for the partial fulfillment of the requirement for the award of the Semester VII- mini Project for degree of Bachelor of Engineering in Information Technology from University of Mumbai.

Examiners

1...

2...

Date:

Place:

CERTIFICATE

This is to certify that the project entitled**“Fire Alarm System with Siren and Water Sprinkler”**is a bonafide work of “**Mohd Salman”(TU4S2021018)**,“**Umaid Shaikh” (TU4S2021005), “Hrushikesh Dunde”(TU4S2021008), “Vivek Shirsat” (TU4F1819070)** submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the degree of “Bachelor ofEngineering”in “Information Technology”

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No project is ever complete without the guidance of those expert who have already traded this past before and hence become master of it and as a result, our leader. So we would like to take this opportunity to take all those individuals how have helped us in visualizing this project.

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ABSTRACT

Fire presents significant threat to life due to its severe hazards and ability to spread rapidly. fire poses a huge threat to human life. Fire detection systems, particularly vision-based systems, identify flames before any loss or destruction occurs. In this model, a novel vision-based technology is created that uses a camera to detect flames (the visible part of a fire) over long distances. An immediate alert is generated on android application. The goal of the proposed system is to notify the remote user when a fire accident occurs. This system can be installed in any remote area where there is a risk of fire. Using this system, we can detect a fire by using a fire sensor beside the camera. After detection of fire the camera is used to take the fire photo and it goes to micro controller. By using camera method, the report is automatically generated and delivered to the person immediately following the fire is detected in any part of the frame using Wi-Fi/GSM. To prevent major damage, the detector will detect flames caused by a fire accident and activate the water sprinkler. The appliance system includes components such as a buzzer for alarming, displaying temperature, humidity and to put out the fire, we use a motor pump.

1.Introduction

Fire is a serious danger to life and property in worldwide. It is usually caused by combustion of materials which releases heat and light in large amount. Fire accident is common feature in factories, house, markets etc. due to inadequate fire protection and a lack of adequate fire alarm system. So we try to design automate fire detection with water sprinkler system because the event is very dangerous in our life A good firefighting system is one that reduces fire damage while also limiting the harm caused by the firefighting system itself. Fires have become a serious issue in recent years, and they must be dealt with quickly and efficiently to avoid the loss of lives and property. When the observed temperature exceeds 50degrees, it is considered a fire situation. It takes about 15 minutes for personnel to arrive for help in the event of fire threats in vital places such as hospitals, schools, and banks. Appropriately allocating fire alarms with proactive warnings could save lives and prevent property loss.

- This system will help in minimizing loss of lives and property.
- It can be time efficient compared to waiting for the fire ambulance.
- The sound can alert the neighboring houses as well and make them alert as well.

A project like this will be of great benefit in every household of the society. In case of a fire, this system will alert everyone in the fire affected area when the siren blows. Also, the water sprinkler will do the work to minimize or put out the fire. Similarly, setting up this system in villages and hard-to-access areas will be of immense help. In case of a large fire, it helps to save lives and provide ample time to call the fire brigades. Although it’s just a start, but it can certainly help to develop certain advances in this field. Engineering is all about solving problems using math, science, and technical knowledge. As this was the first opportunity for us to engage ourselves actually in the electrical engineering, we utilized this at our best. Before creating something, we have properly defined the problem and contributed a little in solving fire problems in each sector possible by creating this project.

2.Literature Survey

[1] D. Teja, M. Suraj Kiran, A.V. Sudheer, I. Anil Kumar Reddy

Fire presents significant threat to life due to its severe hazards and ability to spread rapidly. fire poses a huge threat to human life. Fire detection systems, particularly vision-based systems, identify flames before any loss or destruction occurs. In this model, a novel vision-based technology is created that uses a camera to detect flames (the visible part of a fire) over long distances. An immediate alert is generated on android application. The goal of the proposed system is to notify the remote user when a fire accident occurs. This system can be installed in any remote area where there is a risk of fire. Using this system, we can detect a fire by using a fire sensor beside the camera. After detection of fire the camera is used to take the fire photo and it goes to micro controller. By using camera method, the report is automatically generated and delivered to the person immediately following the fire is detected in any part of the frame using Wi-Fi/GSM. To prevent major damage, the detector will detect flames caused by a fire accident and activate the water sprinkler. The appliance system includes components such as a buzzer for alarming, displaying temperature, humidity and to put out the fire, we use a motor pump.

[2] M. S. Bin Bahrudin, R. A. Kassim and N. Buniyamin

The key feature of the system is the ability to remotely send an alert when a fire is detected. When the presence of smoke is detected, the system will display an image of the room state in a webpage. The system will need the user confirmation to report the event to the Firefighter using Short Message Service (SMS). The advantage of using this system is it will reduce the possibility of false alert reported to the Firefighter. The camera will only capture an image, so this system will consume a little storage and power.

[3] Hari Varshini S, Boomika S, Sherene Amalia G, Leena R

Fire alarm systems are essential in alerting people before fire engulfs their homes. However, fire alarm systems, today, require a lot of wiring and labor to be installed. This discourages users from installing them in their homes. Therefore, we are proposing an IoT based wireless fire alarm system that is easy to install. The proposed system is an ad-hoc network that is distributed over the house. This system consists of a microcontroller (ESP8266 nodeMCU) connected to an infrared flame sensor that continuously senses the surrounding environment to detect the presence of fire. The microcontrollers create their own Wi-Fi network. Once fire is detected by a sensor, it sends a signal to a microcontroller that is triggered to send an SMS to the user, call the user and alert the house by producing a local alarm. The user can also get information about the status of his home via sending an SMS to the system. A prototype was developed for the proposed system and it carried out the desired functionalities successfully with an average delay of less than 30 seconds.

[4] Alqourabah, Hamood & Muneer, Amgad & Fati, Suliman. (2020).

A Smart Fire Detection System using IoT Technology With Automatic Water Sprinkler.. House combustion is one of the main concerns for builders, designers, and property residents. Singular sensors were used for a long time in the event of detection of a fire, but these sensors can not measure the amount of fire to alert the emergency response units. To address this problem, this study aims to implement a smart fire detection system that would not only detect the fire using integrated sensors but also alert property owners, emergency services, and local police stations to protect lives and valuable assets simultaneously. The proposed model in this paper employs different integrated detectors, such as heat, smoke, and flame. The signals from those detectors go through the system algorithm to check the fire's potentiality and then broadcast the predicted result to various parties using GSM modem associated with the system. To get real-life data without putting human lives in danger, an IoT technology has been implemented to provide the fire department with the necessary data. Finally, the main feature of the proposed system is to minimize false alarms, which, in turn, makes this system more reliable. The experimental results showed the superiority of our model in terms of affordability, effectiveness, and responsiveness as the system uses the Ubidots platform, which makes the data exchange faster and reliable.

[5] May Zaw Tun , Htay Myint

The fire detection system combines the simultaneous measurements of smoke, carbon monoxide, and carbon dioxide. The securityof campus against intruders moving in laboratories, class rooms, staffrooms or washrooms. The fire alarm system consists of Firedetectors (with can be smoke detector, heat, or Infra-Red detectors), control unit and alarm system. A fire detection system isdeveloped based on the simultaneous measurements of temperature and smoke. The fire detection system with the alarm algorithmdetected fires that were not alarmed by smoke sensors, and alarmed in shorter times than smoke sensors operating alone.Previous fire detection algorithms used data from sensors for temperature, smoke, and combustion products. The smoke sensoralarms when the analog output signal exceeds or equal the threshold value. The node includes analog sensors to measure smoke,carbon monoxide (CO) and temperature. A fire alarm system should reliably and in a timely way notify building occupants aboutthe presence of fire indicators, such as smoke or high temperatures.

[6] Ashwini C , Delfin.S , Nelluri Harinadh

The fire is a kind of disaster threatening the social wealth and humanity's safety. They cause threats to the residential community and may results in deaths and property damage. In present days, people construct buildings without any proper safety measures and Fire usually occurs in homes because of carelessness and changes in the environmental conditions.so we need a system that can detect the fire accidents. Fire alarm system is needed in different places like buildings, hospitals, libraries, schools and banks etc.so in this paper we proposed a fire alarm system using Arduino. The primary purpose of the system is to provide early warning of fire. So, that the people can be evacuated & immediate action can be taken to eliminate or reduce of the fire.in this system alarm can be trigged by using detectors or by manual cell points.

[7] Zephaniah Shiwalo Obanda

This research aimed at proposing a prototype of a fire detection system using a multi-sensor approach. This research applied rapid prototyping methodology for development of the prototype. Data was collected from secondary sources and experimentation. The prototype used an MQ2 gas sensor, a Grove temperature sensor, a Grove light sensor and an Arduino microcontroller, a GSM and GPS shield. In the event of a fire outbreak, the device will be able to send an SMS alert to the home owner as well as the firefighting department with GPS coordinates of the residence. The prototype recorded 83% success rate and 17% false alarm rate based on 6 test cases of which only one failed.

[8] Mr. K. GIRI BABU, M.RAHUL PRAVEEN, G.SRIHARI, B.BALRAJ GOUD, B.SANDEEP KUMAR

This work is to design and implement a Fire and Gas Detection System with water sprinkler using SMS Feedback. This system makes use of a microcontroller along with sensing circuit which will detect gas leakage and fire and with the help of an alarm system the system gives alert about fire or gas leakage and with the installation of a modems & can be sent to notify the user if there is fire or gas leakage and if the fire occurs the water sprinkler sprinkles water on the affected area to reduce the effect of the fire.

[9] Rashmi Vinod Patil , Sayali Fakira Jadhav , Kaveri Sitaram Kapse

Fire Detection Systems are widely used in various safety and security applications. The major amount of fire starts due to the electric short circuit. It leads to damage to property and loss of life. To minimize the damage caused by fire outbreaks due to electric short circuits an IoT technology is used to control such a kind of risk. Traditional fire detection systems are not that effective and quick to alert the owner about fire, in case no one is present on the location. To overcome this problem in this paper we present the design and development of IoT based Fire Detection System. A system that combines qualities for fire, temperature and smoke detection, sending alert Text Message about the fire to the user along with onsite Buzzer, updating temperature, humidity and smoke on ThingSpeak cloud every 15 seconds, and it also moves manually with the help of Android Application. The Fire Detection System consists of main parts: Multiple sensors, communication system, motion planning and application for manual patrolling of the system. Can be used in college, school, office, and industry for safety purposes.

3.Methodology

A step-by-step design methodology of a typical design project and the topics can be divided into three primary stages as described below:

Stage 1:

- ✓ Identify system's design requirements/specifications.
- ✓ Discuss the design concepts with teammates.
- ✓ Agree on a basic design for each project subsystem.
- ✓ Create subgroups to concentrate on various aspects of the design.

Stage 2:

- ✓ Prepare an anticipated progress chart of the project.
- ✓ Create a parts list. • Purchase or otherwise acquire equipment.
- ✓ Build a prototype of each component.
- ✓ Test the prototype.
- ✓ Build the interface.
- ✓ Test the interface.

Stage 3:

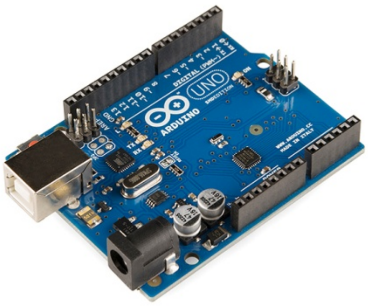
- ✓ Build the system.
- ✓ Test the system.
- ✓ Make final adjustments.
- ✓ Finalize the system.
- ✓ An organized, methodological, effective design process, as described above, can improve quality, reduce costs, and speed time to market, and thereby better matching products to customer needs

Problem Definition

Safety is a crucial consideration in the design of residential and commercial buildings to safeguard against the loss of life and damage to property. The existing fire alarm system on market nowadays is too complex in terms of its design and structure. Since the system is too complex, it needs regular maintenance to be carried out to make sure the system operates well. Meanwhile, when the maintenance is being done to the existing system, it could raise the cost of the system

Hardware Requirements

1. Arduino Uno :



- ❖ Digital I/O Digital inputs/outputs were：0~13Analog I/O Analog input/output: 0~5
- ❖ Support ISP download function.
- ❖ The input voltage: when connected to the computer USB without external power supply.
- ❖ External power supply 7 v to 12 v dc input voltage.
- ❖ Output voltage: 3.3 V and 5 V dc voltage output dc voltage output.

2. Fire Sensor Module :

- ❖ The Matrix-Fire_Sensor module is a flame detector module. It uses the YS-17 flame detector to detect flames or fires.
- ❖ It can detect flames or fires whose wavelengths are between 760nm and 1100nm.
- ❖ Its detecting angle is up to 60 degrees. Its sensitivity can be adjusted via the potentiometer.

3. 5v Relay Module :

- ❖ This module is in line with international safety standard, control areas and load area have the isolation groove; the double FR - 4 circuit board design.
- ❖ high-end SMT process

4. 5V DC Pump :

- ❖ Voltage range (V): DC2.5-6v
- ❖ Rated voltage: DC3V or 4.5V
- ❖ No load of water discharge capacity: 100L / H
- ❖ Load rated current: 0.18A
- ❖ Use: diving type
- ❖ The pump is a DC pump, 3V pump with DC3-4.5V power supply (Can not be used directly with 220V AC voltage)

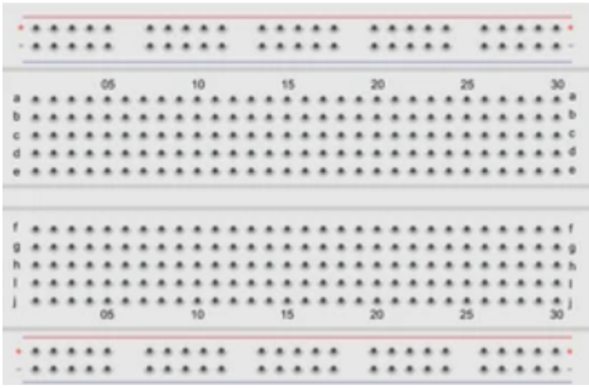
5. Buzzer :

- ❖ They have good characters like small cubage.
- ❖ They have low wastage.
- ❖ They have high voice pressure.
- ❖ 2 Terminals,complete drive circuit incorporated for direct operation from AC power source.

6. 5V AC DC Adapter :

- ❖ An AC-DC power supply or adapter is an electrical device that obtains electricity from a grid-based power supply and converts it into a different current, frequency, and voltage.

7. Breadboard :



- ❖ An electronics breadboard (as opposed to the type on which sandwiches are made) is actually referring to a solderless breadboard.
- ❖ These are great units for making temporary circuits and prototyping, and they require absolutely no soldering.

8. JUMPER WIRES:

- ❖ It is used for connecting the sensors , microcontroller and other components.

SOFTWARE USED

Arduino Uno v1.8.13

Arduino IDE is the **free software used to program Arduino circuit boards**. Arduino IDE has a unique programming language to ensure that all the hardware products associated with it can be programmed in the same way. It's also **open-source**, and so many tech-savvy individuals have taken to creating boards of their own.

Arduino IDE is one of a few programs, such as Arduino Builder, that are compatible with PCs systems. Arduino boards have been used for several creative purposes, including turning on lights and **controlling other electrical appliances**. The possibilities are endless.

How it works

Once you've installed Arduino IDE through a simple installation process, plug in your Arduino board to it as well. You must also ensure that the software attempts to connect to the correct type of Arduino board. There are multiple **demo programs** that will help you get to speed with how to code your board.

Additionally, it's essential to note that your Arduino will have to be **powered by your computer** or, if that isn't available, you can also power it with a battery or an AC-to-DC adapter.

Installing your code

Installation is as simple as using the Arduino software on your Windows computer and then uploading your code to your Arduino using a **USB cable**. There's no limit to what you can do if you know how to code. You can even create a robot.

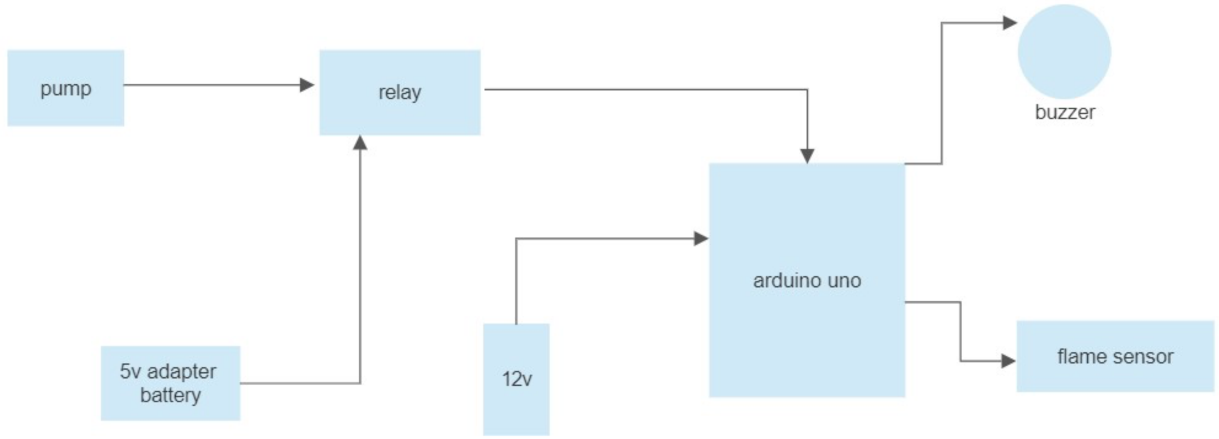
Compatibility

Arduino IDE will **work on any Arduino circuit boards**, including the **Arduino UNO**, which is an affordable option. Of course, this means that you can also use the software on the **Arduino TIAN and Ethernet**, or any boards of this kind. You can download this free open-source software with us.

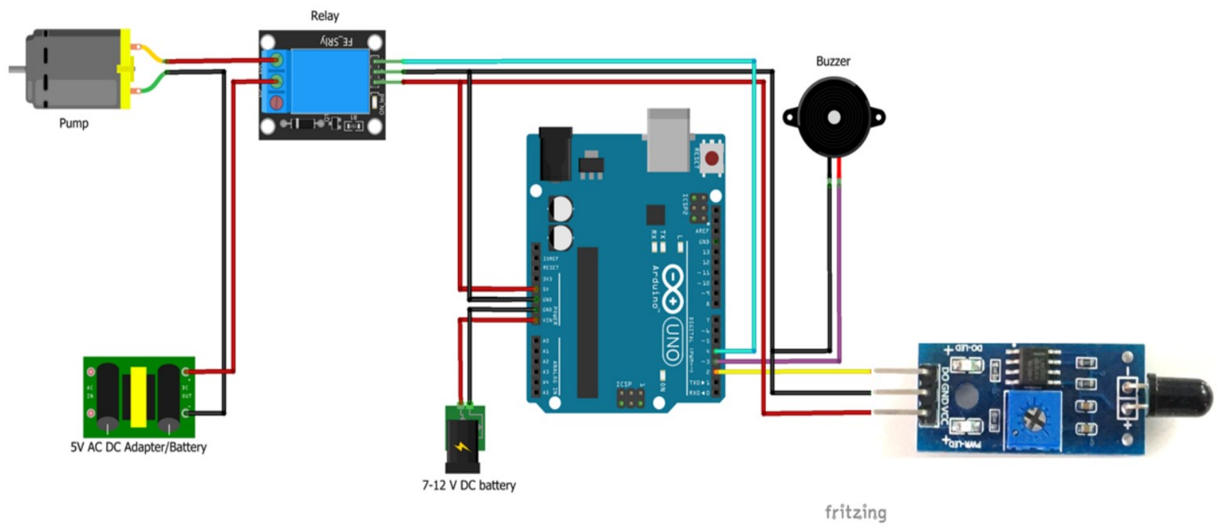
Learning curve

Using Arduino IDE has a learning curve because of its **unique programming language**. That being said, once you get the hang of it, you'll be on your way to programming your Arduino to perform its own unique functions. People have even gone as far as to use it as a garden controller for automatic watering. Do you have any ideas that you want to try out

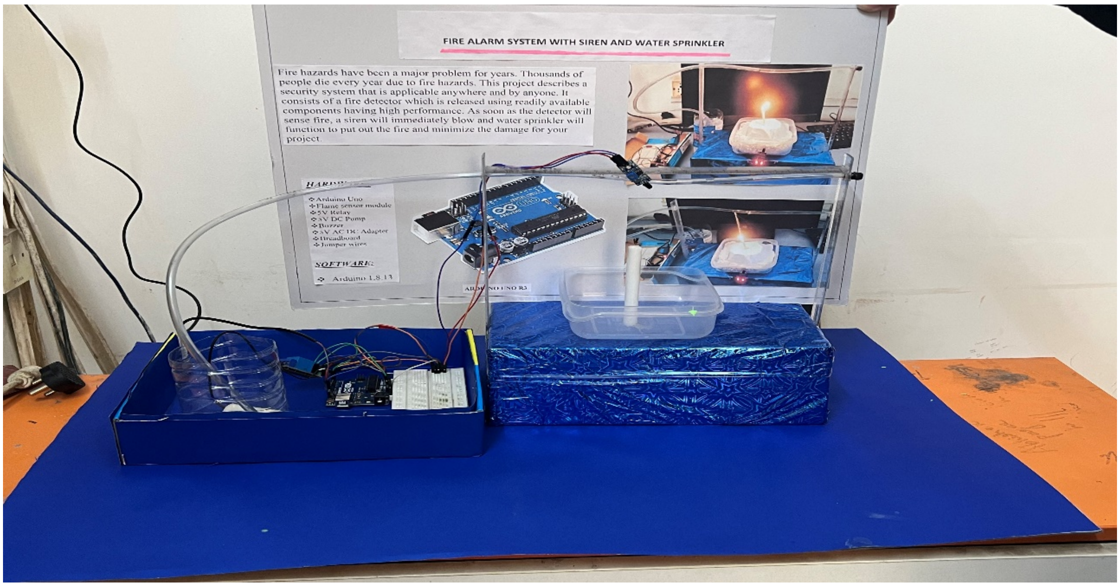
Block diagram :

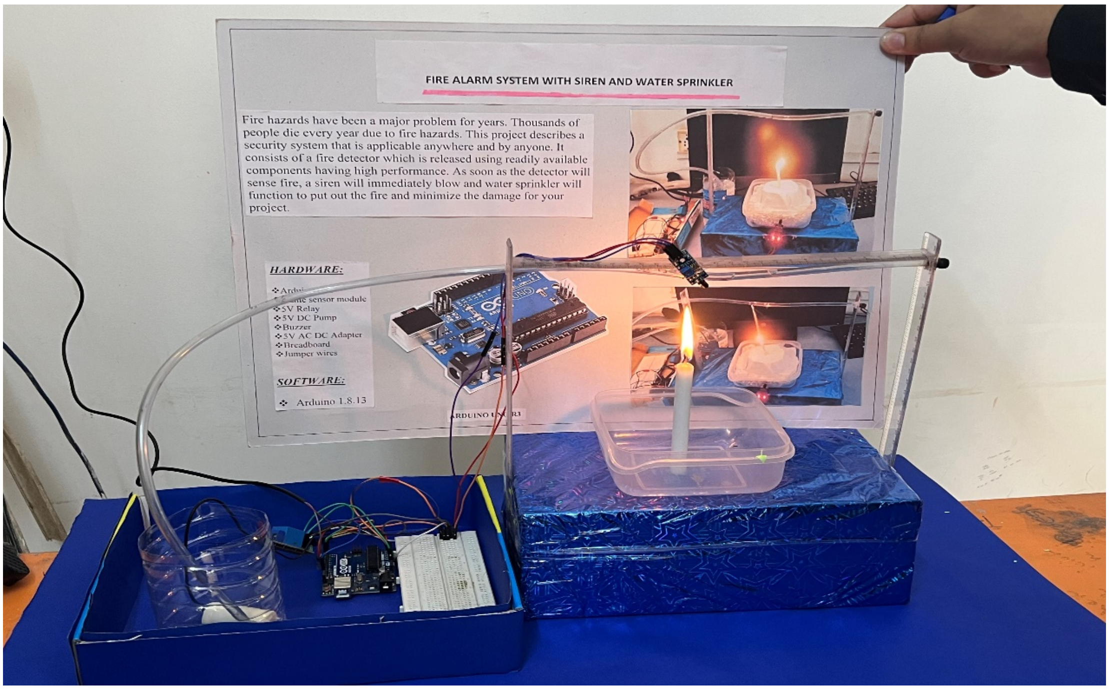


Circuit Diagram :



4. RESULT AND ANALYSIS





CODE

```
#define SENSOR_PIN 2
#define BUZZER_PIN 3
#define RELAY_PIN 4
#define SPRINKLER_START_DELAY 5000
#define SPRINKLER_ON_TIME 3000
unsigned long previousTime = millis();
void setup()
{
 pinMode(RELAY_PIN, OUTPUT);
 pinMode(SENSOR_PIN, INPUT);
}
void loop()
{
 int sensorValve = digitalRead(SENSOR_PIN);
if (sensorValve == LOW)
{
analogWrite(BUZZER_PIN, 50);
if(millis() - previousTime > SPRINKLER_START_DELAY)
{
digitalWrite(RELAY_PIN, LOW);
delay(SPRINKLER_ON_TIME);
}
}
```

```
else
{
analogWrite(BUZZER_PIN, 0);
digitalWrite(RELAY_PIN, HIGH);
previousTime = millis();
}
```

Explanation of the Code

To receive the digital signal from the flame sensor, port A1 is used, which can be changed in the sketch to any general-purpose port;This sensor has a “logical unit” signal on the digital output when an open flame appears up to 1 meter away from the IR receiver and then the buzzer starts when flame is detected .timer of the sprinkler is set to 5 seconds after 5 seconds the water starts sprinkling .

CONCLUSION

We have designed all the sections of our project successfully. Initially, the design ofthe circuit was made and it was tested on breadboard. The significance and need for this project has already been thoroughly discussed.Although some problems and errors were encountered during the progress of the project we were able to deal with them successfully making our project as effective as possible. We want to express our sincere gratitude to our supervisor, teachers and seniors for their great support throughout the completion of the project.

6.REFERENCES

[1] Bu, F. and Gharajeh, M.S., 2019. Intelligent and vision-based fire detection systems: A survey. Image and VisionComputing,91, p.103803.

[2] Saeed, F., Paul, A., Rehman, A., Hong, W.H. and Seo, H., 2018. IoT-based intelligent modeling of smart homeenvironmentforfire preventionandsafety.JournalofSensorandActuatorNetworks,7(1),p.11.

[3] Saeed, F., Paul, A., Karthigaikumar, P. and Nayyar, A., 2019. Convolutional neural network based early firedetection.MultimediaToolsandApplications, pp.1- 17.

[4] Kodur,V.,Kumar,P.andRafi,M.M.,2019.Firehazardinb uildings:review,assessmentandstrategiesforimproving firesafety.PSU ResearchReview..

[5] Mahzan, N.N., Enzai, N.M., Zin, N.M. and Noh, K.S.S.K.M., 2018, June. Design of an Arduino-based home firealarm system with GSM module. In Journal of Physics: Conference Series (Vol. 1019, No. 1, p. 012079). IOPPublishing.

[6] Suresh,S.,Yuthika,S.andVardhini,G.A.,2016,Novemb er.Homebasedfiremonitoringandwarningsystem.In20 16InternationalConferenceonICTinBusinessIndustry &Government(ICTBIG)(pp.1-6).IEEE.

[7] Kanwal, K., Liaquat, A., Mughal, M., Abbasi, A.R. and Aamir, M., 2017. Towards development of a low cost earlyfire detection system using wireless sensor network and machine vision. Wireless Personal Communications, 95(2),pp.475-4

[8] Nfpa.org.NFPAData,Research,AndTools.[online]Available:https://w ww.nfpa.org/News-and-Research/Data-research-andtools2018

[9] Liu,Z.“Reviewof Recent Developmentsin Fire Detection Technologies”. Journal of Fire Protection Engineering,13(2), pp.129-151,2003.

[10] Mowrer,F.” Lagtimesassociated with fire detection and suppression”.Fire Technology,26(3),pp.244- 265,2010.